Advanced Statistics

29-01-26

Uniform dist

PDF → 2 parameters

theory

a - min

b - max

python

loc = min

scale = max - min

eg.

loc = 1 , scale = 3

min = a = 1

max - min = 3

b - a = 3

b - 1 = 3

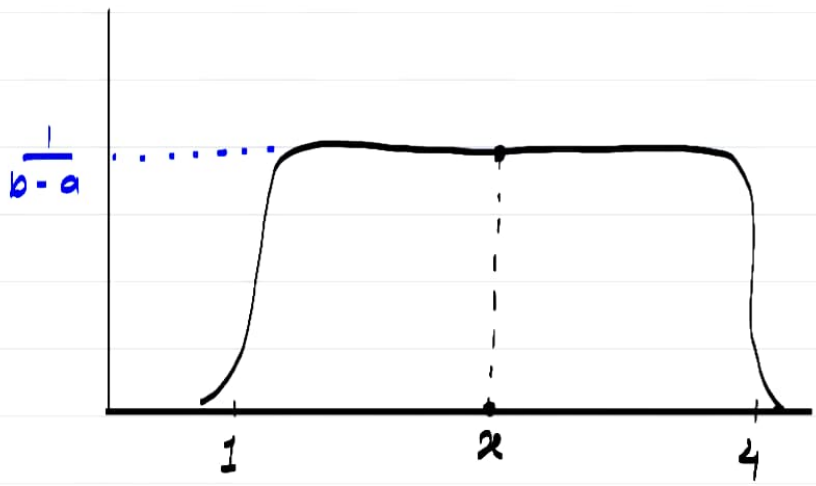
b = 4

PDF: $f(x) = \frac{1}{b-a}$ if $a \le x \le b$

0 otherwise

a = 1 b = 4

min max



Important points about Normal distribution

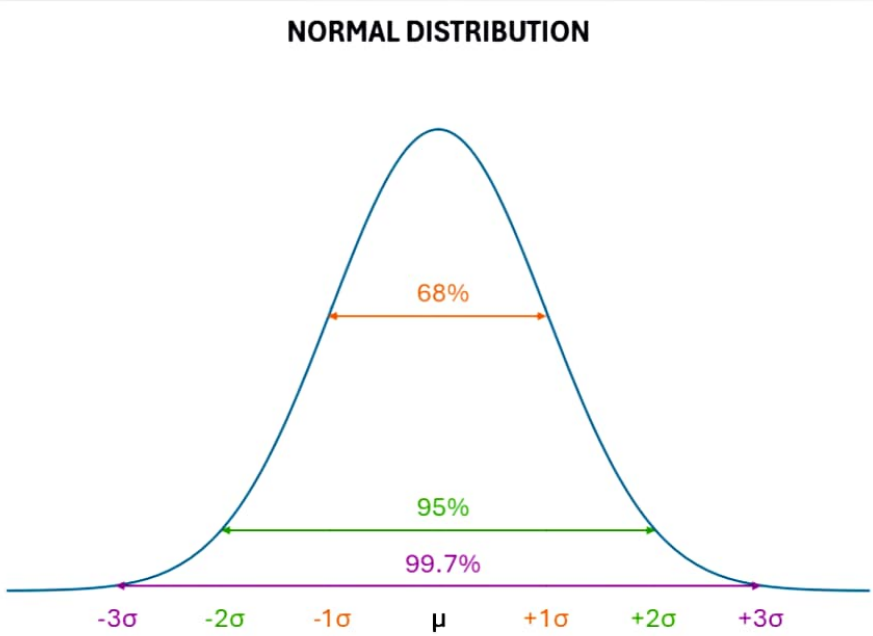
- It models various natural phenomena
- very important in Inferential statistics (central limit theorem)
- very useful in machine learning, deep learning
- very useful in interpreting statistical results.

Six sigma limits rule:

Normal distribution has 2 parameters - mean and std dev

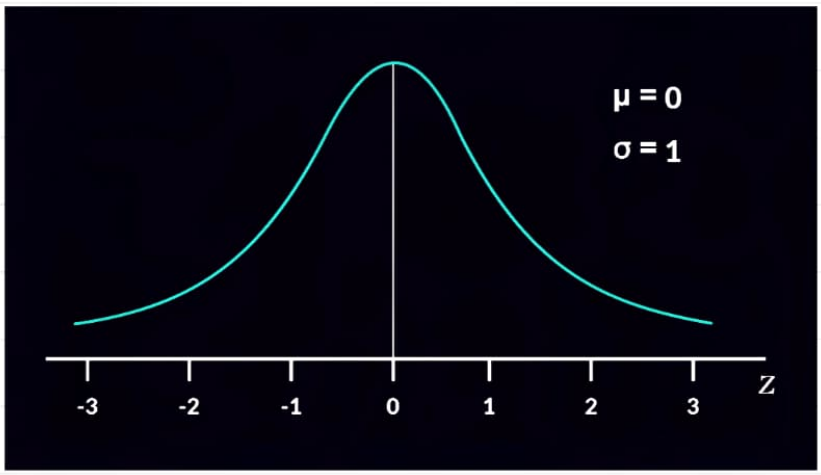
mean = μ
std dev = σ

Range	proportion of data	
$\mu \pm \sigma$	68%	1 sigma limit
$\mu \pm 2\sigma$	95%	2 sigma limit
$\mu \pm 3\sigma$	99.7%	3 sigma limit



Standard Normal dist

- Its a special case of normal dist having
  mean = 0
  std dev = 1



- Any normal dist can be converted to standard normal distribution. This technique is called standardization and its a very popular data preprocessing technique in machine learning.

lets say

data = [x_1 x_2 x_3 ..... x_n]
follows normal distribution

$$\mu = \frac{\sum_{i=1}^{n} x_i}{n} \qquad \sigma = \left(\frac{\sum_{i=1}^{n} (x-\mu)^2}{N} \right)^{1/2}$$

$$\text{data new} = \left[\frac{x_1-\mu}{\sigma}, \frac{x_2-\mu}{\sigma}, \frac{x_3-\mu}{\sigma} \ldots\ldots \frac{x_n-\mu}{\sigma} \right]$$

follow standard normal distribution

standardization

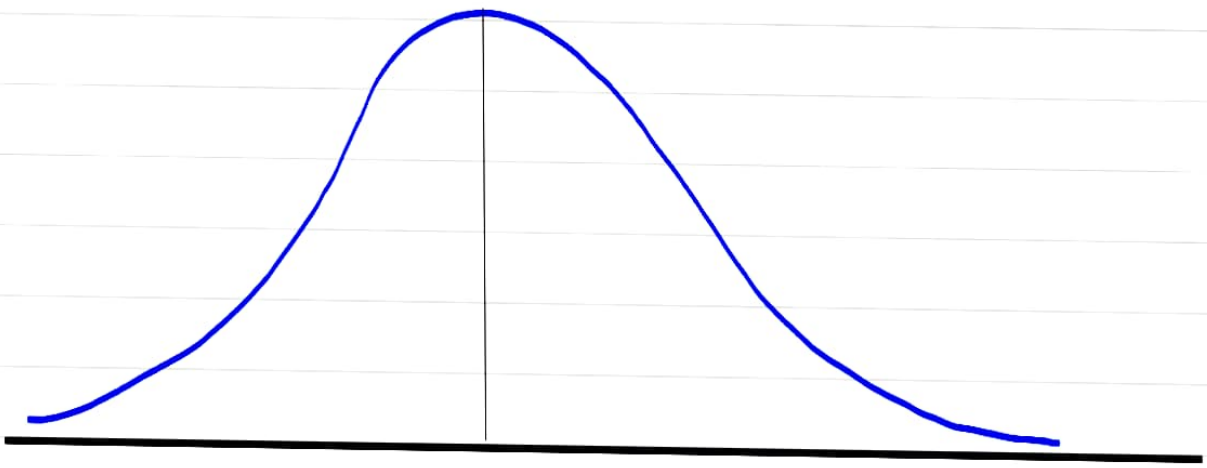
Test of Normality

checking whether given data follows normal dist or not.

1) Visual and non-visual observation

- plot the KDE of given data. if KDE looks symmetrical and bell shaped then its a normal dist

- compute skewness and kurtosis
  if skewness ≈ 0 and kurtosis ≈ 3 then its a normal dist



mean = median = mode perfectly normal dist

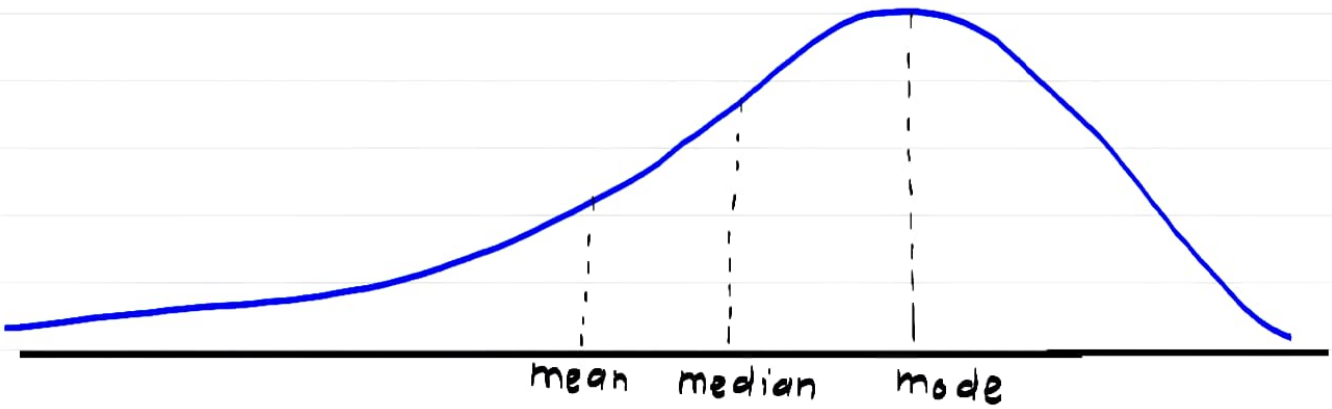
mean $\approx$ median $\approx$ mode nearly normal dist

- compute mean, median, mode. if
  mean $\approx$ median $\approx$ mode then its a normal dist.

right skewed data

mode < median < mean

left skewed data



mean < median < mode

- we can also use six-sigma limits to find if the distribution is normal.

Q Q plot test for normality check

quantile - quantile plot

data = $[x_1 \; x_2 \; x_3 \ldots\ldots x_n]$

find all percentile values of this given data

percentile data = $[P_1 \; P_2 \; P_3 \; P_4 \ldots - P_{100}]$

P_i = i^{th} %le

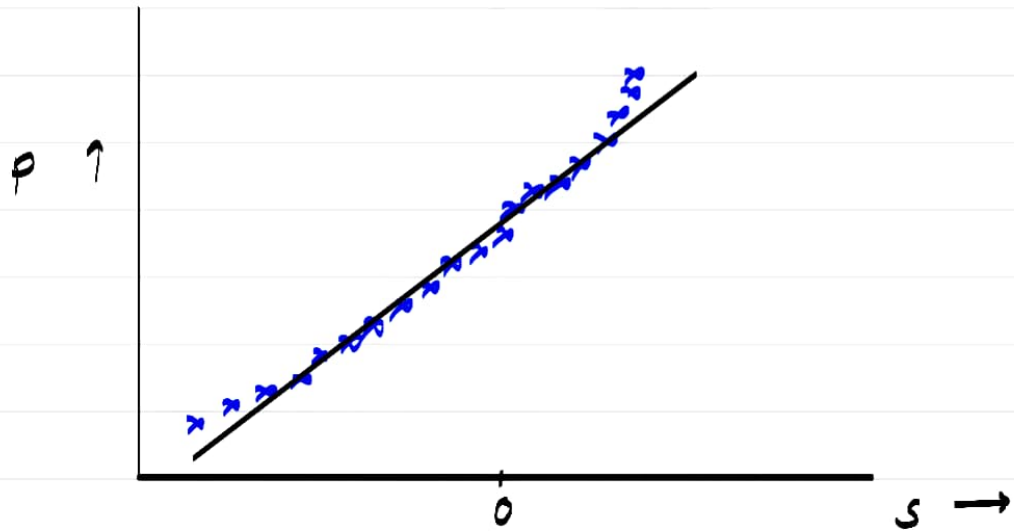
P_{50} = 50th %le = median

percentile points from standard normal distribution
= $[S_1 \; S_2 \; S_3 \ldots - S_{100}]$

form the pairs of percentiles from both data

pairs = $[P_1 S_1, \; P_2 S_2, \; P_3 S_3, \; P_4 S_4 \ldots\ldots P_{100} S_{100}]$

plot these pairs on a scatter plot
↓
bivariate analysis



If the scatterplot between quantiles of data and quantiles of standard normal dist show the points lying on a straight line then we can say that given data follows normal distribution.

- Data is not normally distributed if scatterplot doesn't follow a straight line.

Data Transformation

- if data is not normally distributed, then we can transform it and try to see if transformed data follows normal dist.

- We might need this data for ML, statistical tests where normality of data is very important.

1) Log transformation

- apply logarithm on the data
- works well for skewed dataset

$$x' = \log(x)$$

- mostly used when all datapoints are positive